

Exercise 1 : 10 pts

- 1 Using the truth table show that the logical implication is a transitive connective, that is :
 $((P \Rightarrow Q) \wedge (Q \Rightarrow R)) \Rightarrow (P \Rightarrow R)$.

1.5 pts

P	Q	R	$P \Rightarrow Q$	$Q \Rightarrow R$	$P \Rightarrow R$	$(P \Rightarrow Q) \wedge (Q \Rightarrow R)$	Final Implication
1	1	1	1	1	1	1	1
1	1	0	1	0	0	0	1
1	0	1	0	1	1	0	1
1	0	0	0	1	0	0	1
0	1	1	1	1	1	1	1
0	1	0	1	0	1	0	1
0	0	1	1	1	1	1	1
0	0	0	1	1	1	1	1

- 2 For any natural number n :

- a Expand the expression : $(n-1)n(n+1)(n^2+1)$

0.5 pts

$$= n(n^2-1)(n^2+1) = n(n^4-1) = n^5 - n.$$

- b Prove that : 5 divides $(n^5 - n)$.

2 pts

Proof by cases :

$$\forall n \in \mathbb{N} : (n = 5k) \vee (n = 5k+1) \vee (n = 5k+2) \dots\dots\dots 0.5$$

$$n = 5k \text{ means } 5 \text{ divides } n \text{ which is a factor of } n^5 - n \dots\dots\dots 0.5$$

$$n = 5k+1 \Rightarrow n^2-1 = 5(5k^2+2k) // \dots\dots\dots 0.5$$

$$n = 5k+2 \Rightarrow n^2+1 = 5(5k^2+4k+1) // \dots\dots\dots 0.5$$

One can also proceed as follows : $\forall n \in \mathbb{N} : (n = 5k) \vee (n = 5k+1) \vee (n = 5k+2) \vee (n = 5k+3) \vee (n = 5k+4) \dots$

$$\text{Or by induction : } (n+1)^5 - (n+1) = (n^5 - n) + 5(n^4 + 2n^3 + 2n^2 + n).$$

- 3 Let S be a finite set with $|S| = n \in \mathbb{N}$. Prove by induction that $|\mathcal{P}(S)| = 2^n$ ($\mathcal{P}$ denotes the power set).

2 pts

$$\text{Base case : } n = 0 \text{ means } S = \emptyset \text{ then } |\mathcal{P}(\emptyset)| = |\{\emptyset\}| = 1 = 2^0. \text{ checked.} \dots\dots\dots 0.5$$

$$\text{Inductive step : Assumption : } |S| = n \Rightarrow |\mathcal{P}(S)| = 2^n.$$

$$\text{Let } S' \text{ such that } |S'| = n+1 \text{ and consider its partition : } S' = (S' \setminus \{x\}) \cup \{x\} \text{ where } x \in S'.$$

$$|S' \setminus \{x\}| = n, \text{ so from the induction assumption : } S' \setminus \{x\} \text{ has } 2^n \text{ subsets.}$$

$$\text{Since } \mathcal{P}(S') \text{ contains all subsets of } S' \setminus \{x\} \text{ and the same subsets adjoined } x, \text{ we can say it has : } 2^n + 2^n = 2^{n+1} \text{ subsets.} \dots\dots\dots 1.5$$

Proofs without induction are not accepted.

4 We define on $\mathbb{R}^2$ the binary relation $\mathfrak{R}$ by : $((x, y) \mathfrak{R} (x', y')) \Leftrightarrow (\exists \alpha \in \mathbb{R}^* : (\alpha x, \alpha^2 y) = (x', y'))$.

a Show that $\mathfrak{R}$ is an equivalence relation.

2 pts

Reflexivity : $\alpha = 1$ 0.5

Symmetry : $(\alpha x, \alpha^2 y) = (x', y') \xRightarrow{\alpha \neq 0} \left(\frac{1}{\alpha} x', \left(\frac{1}{\alpha} \right)^2 y' \right) = (x, y)$ 0.5

Transitivity : $\begin{cases} (\alpha x, \alpha^2 y) = (x', y') \\ (\beta x', \beta^2 y') = (x'', y'') \end{cases} \Rightarrow ((\alpha\beta)x, (\alpha\beta)^2 y) = (x'', y''). (\alpha\beta \neq 0) \dots 0.5$

Conclusion 0.5

b Given $(a, b) \in \mathbb{R}^2$, define $\overline{(a, b)}$ the class of equivalence of (a, b) ?

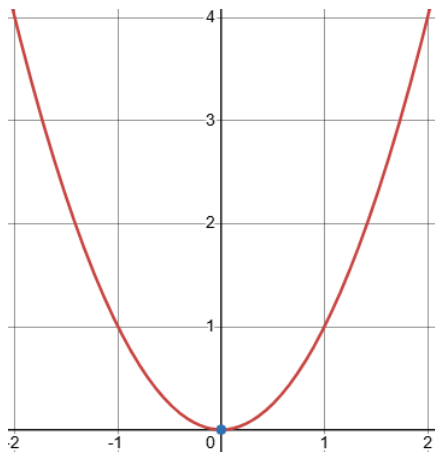
0.5 pts

$$\overline{(a, b)} = \{(\alpha a, \alpha^2 b) \in \mathbb{R}^2 / \alpha \in \mathbb{R}^*\}$$

c Determine and represent graphically $\overline{(1, 1)}$.

1.5 pts

$\{(\alpha, \alpha^2) \in \mathbb{R}^2 / \alpha \in \mathbb{R}^*\}$ is the graph of equation $y = x^2$ except the origin 1



.....0.5

Exercise 2 : 10 pts

$e \simeq 2,718281828...$ is the base of the natural logarithm

1 Consider the function $f : \mathbb{R} \rightarrow]-1, 1[$ defined by $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$.

a Show that : $\forall x \in \mathbb{R} : f(x) = 1 - \frac{2}{e^{2x} + 1}$.

0.5 pts

$$\frac{(e^x - e^{-x})e^x}{(e^x + e^{-x})e^x} = \frac{e^{2x} - 1}{e^{2x} + 1} = 1 - \frac{2}{e^{2x} + 1}.$$

b Is f injective, surjective? justify using definitions.

2.5 pts

f is injective : $f(x) = f(x') \Rightarrow 1 - \frac{2}{e^{2x} + 1} = 1 - \frac{2}{e^{2x'} + 1} \Rightarrow \frac{2}{e^{2x} + 1} = \frac{2}{e^{2x'} + 1} \Rightarrow e^{2x} = e^{2x'} \Rightarrow x = x'$1
 f is surjective : $\forall y \in]-1, 1[: f(x) = y \Leftrightarrow 1 - \frac{2}{e^{2x} + 1} = y \Leftrightarrow \frac{2}{e^{2x} + 1} = 1 - y \xLeftrightarrow[y \neq 1] \frac{e^{2x} + 1}{2} = \frac{1}{1 - y} \Leftrightarrow e^{2x} = \frac{2}{1 - y} - 1$. Since $y > -1$ then $1 - y < 2$ and $\frac{2}{1 - y} - 1 > 0$. So we can apply the logarithm to get : $x = \frac{1}{2} \ln \left(\frac{2}{1 - y} - 1 \right) \in \mathbb{R}$1.5

2 Let $G =]-1, 1[$ and $a \in G$. Knowing that the function

$$\varphi : [-1, 1] \rightarrow \mathbb{R}$$

$$\varphi(x) = \frac{x + a}{ax + 1}$$

is strictly increasing

a Prove that G is closed under the binary operation $\star$ defined by $a \star b = \frac{a + b}{ab + 1}$.

1 pts

$$-1 < b < 1 \Rightarrow \varphi(-1) < \varphi(b) < \varphi(1) \Rightarrow -1 < a \star b < 1.$$

b Establish that $(G, \star)$ is an abelian group.

3 pts

Associativity : $(a \star b) \star c = \frac{a + b + c + abc}{ab + bc + ca + 1} = a \star (b \star c)$1
 Commutativity : $\frac{a + b}{ab + 1} = \frac{b + a}{ba + 1}$0.5
 Identity : $\varphi(x) = a \Rightarrow x = 0$. So 0 is the identity element.0.5
 Inverse : $\varphi(x) = 0 \Rightarrow x = -a$. So $-a$ is the inverse element of a0.5
 Conclusion0.5

3 Show that f is a group isomorphism from $(\mathbb{R}, +)$ to $(G, \star)$.

3 pts

We know that f is bijective from question 1,0.5
 so it is sufficient to prove that it is a group homomorphism :
 $f(a) \star f(b) = \frac{f(a) + f(b)}{f(a)f(b) + 1}$.
 Numerator : $f(a) + f(b) = 2 - \frac{2}{e^{2a} + 1} - \frac{2}{e^{2b} + 1}$.
 Denominator : $f(a)f(b) + 1 = 2 - \frac{2}{e^{2a} + 1} - \frac{2}{e^{2b} + 1} + \frac{4}{(e^{2a} + 1)(e^{2b} + 1)}$.
 Add and subtract $\frac{4}{(e^{2a} + 1)(e^{2b} + 1)}$ to the numerator, extract 1, then reduce the rest to the same denominator :
 $f(a) \star f(b) = 1 - \frac{2}{e^{2(a+b)} + 1} = f(a + b)$2.5